Honors Physics 11 — Formula Cheat Sheet

Procadamia | All formulas, units, and constants

KINEMATICS (UNITS 1–2)

Quantity	Formula	Missing
Velocity	$v = \Delta x/\Delta t$	
Acceleration	$a = \Delta v/\Delta t$	
Kinematic 1	$v = v_0 + at$	Δx
Kinematic 2	$\Delta x = v_0 t + \frac{1}{2}at^2$	v
Kinematic 3	$v^2 = v_0^2 + 2a\Delta x$	t
Kinematic 4	$\Delta x = \frac{1}{2}(v+v_0)t$	a
Free Fall (g)	$g = 9.8\ m/s^2 \downarrow$	

Identify the unknown, pick the equation missing that variable

VECTORS & PROJECTILE MOTION (UNIT 3)

Formula	Notes
$V_x = V \cdot \cos\theta$	Horizontal component
$V_Y = V \cdot \sin\theta$	Vertical component
$V = \sqrt{(V_x^2+V_Y^2)}$	Resultant magnitude
$\theta = \tan^{-1}(V_Y/V_x)$	Direction angle

Projectile Rules

- Horizontal: constant velocity, a=0 → $x = V_x t$
- Vertical: free fall → use kinematics with a=−9.8 m/s²
- At peak: $V_Y = 0$
- Horizontal & vertical are independent

FORCES & NEWTON'S LAWS (UNIT 4)

Law	Formula/Statement
1st (Inertia)	Object at rest stays at rest; in motion stays in motion (unless net force acts)
2nd	$F = ma$ | $a = \Sigma F/m$
3rd (Action-Reaction)	Every action has equal & opposite reaction
Weight	$W = mg$
Friction (kinetic)	$F_k = \mu_k \cdot F_n$
Friction (static max)	$F_s(max) = \mu_s \cdot F_n$
Normal Force	$F_n = mg$ (on flat surface)

Free body diagram first — identify ALL forces before writing ΣF=ma

WORK, ENERGY & POWER (UNIT 5)

Quantity	Formula	Unit
Work	$W = Fd \cdot \cos\theta$	Joule (J)
KE	$KE = \frac{1}{2}mv^2$	J
GPE	$PE = mgh$	J
Elastic PE	$PE = \frac{1}{2}kx^2$	J
Work-Energy	$W_net = \Delta KE$	
Conservation	$KE_1+PE_1 = KE_2+PE_2$	(no friction)
Power	$P = W/t = Fv$	Watt (W)

MOMENTUM & IMPULSE (UNIT 6)

Quantity	Formula	Unit
Momentum	$p = mv$	kg·m/s
Impulse	$J = F\Delta t = \Delta p$	N·s
Conservation	$p_{1i} + p_{2i} = p_{1f} + p_{2f}$	

Collision Type	KE Conserved?	Formula
Elastic	✅ Yes	Objects bounce apart
Inelastic	❌ No	KE lost to heat/sound
Perfectly Inelastic	❌ No	$m_1v_1+m_2v_2=(m_1+m_2)v_f$

ROTATIONAL MOTION & EQUILIBRIUM (UNIT 7)

Quantity	Formula	Unit
Torque	$\tau = r \cdot F \cdot \sin\theta$	N·m
Equilibrium	$\Sigma\tau = 0$ and $\Sigma F = 0$	
Arc length	$s = r\theta$	θ in radians
Tangential v	$v = r\omega$	ω = rad/s
Centripetal a	$a^c = v^2/r = \omega^2 r$	

Equilibrium: sum of CCW torques = sum of CW torques about any pivot

GRAVITY & UNIFORM CIRCULAR MOTION (UNITS 8–9)

Quantity	Formula
Universal Gravity	$F_g = G \cdot m_1 m_2 / r^2$
G constant	$6.674 \times 10^{-11}\ N \cdot m^2/kg^2$
Gravitational field	$g = GM/r^2$
Orbital velocity	$v = \sqrt{(GM/r)}$
Centripetal force	$F^c = mv^2/r = m\omega^2 r$
Period (orbit)	$T = 2\pi r/v$
Kepler 3rd Law	$T^2 \propto r^3\ (T^2/r^3 = const)$

FLUIDS (UNIT 10)

Quantity	Formula	Unit
Density	$\varrho = m/V$	kg/m^3
Pressure	$P = F/A$	$Pa\ (N/m^2)$
Hydrostatic P	$P = P_0 + \varrho gh$	
Buoyant Force	$F_a = \varrho_fluid \cdot g \cdot V_sub$	N
Pascal's Law	$F_1/A_1 = F_2/A_2$	
Continuity	$A_1 v_1 = A_2 v_2$	
Bernoulli	$P + \frac{1}{2}\varrho v^2 + \varrho gh = const$	
Torricelli	$v = \sqrt{(2gh)}$	

> Object floats if ρ_obj < ρ_fluid; sinks if ρ_obj > ρ_fluid

CONSTANTS, UNITS & PROBLEM-SOLVING TIPS

Constants		SI Units	
g (Earth)	$9.8\ m/s^2$	Force	Newton ($N = kg \cdot m/s^2$)
G	$6.674 \times 10^{-11}\ N \cdot m^2/kg^2$	Energy	Joule ($J = N \cdot m$)
ρ_water	$1000\ kg/m^3$	Power	Watt ($W = J/s$)
P_atm	$101{,}325\ Pa \approx 1\ atm$	Pressure	Pascal ($Pa = N/m^2$)

- Always draw a free body diagram (FBD)
- Define positive direction before solving
- Check units in every answer
- Velocity is a vector — include direction
- At peak of projectile: $V_y = 0$ (not v = 0!)
- Energy conservation: no work by friction
- Impulse-momentum: large force × short time = small force × long time